Field: A Ring R with at least two elements is called a field if it is (i) commutative, has (ii) unity (iii) is such that each non-zero element posses a multiplicative inverse.

For example the rational numbers $(\mathbb{Q}, +, \cdot)$ is a field since it is commutative ring with unity and each non zero element is invertiable.

The rings of real numbers and complex numbers are also examples of fields.

Internal Composition.

Let V be any non empty set. if $a*b \in V \ \forall a \in V, b \in V$ and $a*b$ is unique then $*$ said to be an internal Composition.

Let V be any non-empty set F be other non empty set i.e, V and F be any two non-empty sets if $a \circ \alpha \in V \ \forall a \in F$ and $\alpha \in V$ and $a \circ \alpha$ is unique then $\circ$ is said to be an external composition in V over F. Here a is an element of set F and α is an element of V and resultant element $a \circ \alpha$ is an element of set V.

Vector Spaces

Let V be a arbitrary non-empty set objects whose elements are called Vectors. Let F be any set whose elements are called scalars where $(F, +, \cdot)$ is a field.

The set V is said to be vector space if

(i) Closure under addition:

$u + v \in V \quad \forall\ u, v \in V$

(ii) Associative $u, v, w \in V$ show that

$(u+v)+w = u+(v+w)$.

(iii) Commutative $u, v \in V$

$u + v = v + u. \quad \forall\ u, v \in V$

(iv) Existence of additive identity.

there exists a $\bar{0} \in V$ such that $0 + \bar{v} = \bar{v} + 0 = \bar{v}$

$\forall\ \bar{0} \in V$

(v) Existance of additive inverse.

$\bar{u}, (-\bar{u}) \in V$ then $u + (-u) = \bar{0}$

(vi) Closure under scalar multiplication i.e., an external composition V over F

~~$\alpha \in V$~~ $a\alpha \in V \quad \forall\ a \in F$ and $\alpha \in V$

(vii) with internal and external composition it should satisfies.

Distributivity over F $(a+b)\alpha = a\alpha + b\alpha \quad a, b \in F,\ \alpha \in V$ addition.

Di " " " $a(\alpha+\beta) = a\alpha + a\beta \quad a \in F \quad \alpha, \beta \in V$ scalar multiplication

Associativity scalar multiplication. $(ab)\alpha = a(b\alpha) \quad a, b \in F,\ \alpha \in V$

$1 \cdot \alpha = \alpha$. multiplicative Identity.

If V over F not satisfies any one of the above condition of vector space then we say that V is not vector over F.

In other words V is said to be a vector space over the field F

1. $(V, +)$ is an abelian group
2. there is defined an external composition over F in which $a\alpha \in V$ $\forall$ $a \in F$ and ~~$\alpha \in V$~~. $\alpha \in V$
3. and above two compositions it satisfies.

$a(\alpha+\beta) = a\alpha + a\beta$ $\quad \forall\ a \in F,\ \alpha, \beta \in V$

$(a+b)\alpha = a\alpha + b\alpha$ $\quad \forall\ a, b \in F,\ \alpha \in V$

$(ab)\alpha = a(b\alpha)$ $\quad \forall\ a, b \in F$ and $\alpha \in V$

$1 \cdot \alpha = \alpha$. $\quad \forall\ 1 \in F$ and $\alpha \in V$.

Null Space:

The Vector space having only on zero vector $\bar{0}$ is called a zero vector space or null space.

It is clearly satisfies all the postulates of field of scalars.

(i) ~~$a\bar{0} = \bar{0}$ $\forall a \in F$~~, ~~0α~~

Prob: Show that $\mathbb{R}^n$ is a vector space over $\mathbb{R}$.

Sol: Let $x=(x_1, x_2 \cdots x_n) \in \mathbb{R}^n$ $y \in (y_1, y_2 \cdots y_n) \in \mathbb{R}^n$.

Here vector addition $(x+y) = (x_1+x_2 \cdots x_n) + (y_1+y_2+y_3 \cdots y_n)$

$= (x_1+y_1, x_2+y_2, x_3+y_3, \cdots x_n+y_n) \in \mathbb{R}^n$.

Vector addition exists internal composition.

- $\bar{0} \in (0,0,0 \cdots 0) \in \mathbb{R}$ Identity of vector property

Scalar multiplication let $a \in F$ $x \in (x_1, x_2 \cdots x_n) \in \mathbb{R}^n$

then $a \cdot x = (ax_1, ax_2 \cdots ax_n) \in \mathbb{R}^n$.

Scalar multiplication also exist external composition

$\mathbb{R}^n$ satisfies the following postulates of vector space.

1. closure already proved
2. Commutative $(x+y) = (y+x)$

   i.e., $(x_1+y_1, x_2+y_2, x_3+y_3 \cdots x_n+y_n) \in \mathbb{R}^n$
3. Associative $(x+y)+z = x+(y+z)$.

   Ily $(x_1+y_1+z_1, x_2+y_2+z_2 \cdots x_n+y_n+z_n) \in \mathbb{R}$.
4. Identity: $\exists$ a $\bar{0} \in \mathbb{R}^n$

   $0+x = x \in \mathbb{R}^n$.

Existence of additive Inverse

i.e $x \in \mathbb{R}^n$.

s.t $x + (-x) = \bar{0}$ identity (w.r.t internal composition) element.

Closure Under Scal multiplication

i.e., let $a \in \mathbb{R}$ and $x = (x_1, x_2 \cdots x_n) \in \mathbb{R}^n$

then $(ax_1\ ax_2 \cdots ax_n) \in \mathbb{R}^n$.

Distributive of Scalar multiplication over additions

let $a \in \mathbb{R}$ and $X = (x_1, x_2 \cdots x_n) \in \mathbb{R}^n$.

$y = (y_1, y_2 \cdots y_n) \in \mathbb{R}^n$.

i.e., $a(x+y) = ax + ay$

$a(x_1+y_1, x_2+y_2 \cdots x_n+y_n) = (a(x_1+y_1), a(x_2+y_2) \cdots a(x_n+y_n))$

$= (ax_1 + ax_2\ ax_3 \cdots ax_n) + (ay_1 + ay_2 \cdots ay_n)$

$= a(x_1 + x_2 + \cdots x_n) + a(y_1 + y_2 \cdots y_n)$

$a(x+y) = ax + ay \in \mathbb{R}$.

~~$= a(x+y)$~~

Distributive over field addition.

i.e., $(a+b)x = ax + by$.

Now let $a, b \in \mathbb{R}$ $x = (x_1, x_2 \cdots x_n) \in \mathbb{R}^n$.

$(a+b)x = (a+b)(x_1 + x_2 \cdots x_n)$

$= (ax_1 + ax_2 \cdots ax_n) + b \cdots$

$= (a+b)x_1 + (a+b)x_2 + \cdots (a+b)x_n$.

$= (ax_1 + bx_1) + (ax_2 + bx_2) + \cdots (ax_n + bx_n)$.

$= (ax_1 + ax_2 + \cdots ax_n) + (bx_1 + bx_2 + \cdots bx_n)$

$= a(x_1 + x_2 \cdots x_n) + b(x_1 + x_2 + \cdots x_n)$

$= ax + by$.

9. Associative of scales multiplication i.e.,

$$a(bx) = (ab)x.$$

Now $a, b \in \mathbb{R}$ $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

$$bx = b(x_1, x_2, \ldots, x_n)$$

$$= (bx_1, bx_2, \ldots, bx_n)$$

$$a(bx) = a(bx_1, bx_2, \ldots, bx_n)$$

$$= (abx_1, abx_2, \ldots, abx_n)$$

$$a(bx) = (ab)(x_1, x_2, \ldots, x_n)$$

10. Multiplicative Identity i.e $1 \cdot x = x$.

Let $1 \in \mathbb{R}$ ~~$x \in$~~ $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

Now $1 \cdot x = 1 \cdot (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

$$= (1 \cdot x_1, 1 \cdot x_2, \ldots, 1 \cdot x_n) \in \mathbb{R}^n.$$

$$= (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n.$$

$$= x.$$

Hence $\mathbb{R}^n$ is a Vector over the field $\mathbb{R}$.

§. S.t. Theset C_n of all - n-tupples of complex numbers ~~as the~~ with addition and multiplication is the external composition and scalar-multiplication of complex numbers by complex number plane is a Vector space over the field of Complex Complex numbers.

Prob:

The set $V = \{(x,y) \in \mathbb{R}^2 \mid x+y=0\}$ is a vector space?

internal composition $(x_1,y_1)+(x_2,y_2) = (x_1+x_2, y_1+y_2)$

$= (x_1+x_2+y_1+y_2)$ $\quad x_1, x_2 \in \mathbb{R}^2$, $y_1, y_2 \in \mathbb{R}^2$

$= 0+0 \in V$

Scalar Multiplication let $a \in R$ and $(x,y) \in V$

then $a(x+y) = (ax, ay) = ax+ay = a(x+y)$

$= a\bar{0} = 0 \in \bar{V}$.

both internal and external compositions exist.

other properties left for exercise.

Hence $V = \{(x,y) \in \mathbb{R}^2 \mid x+y=0\}$ over R is a vector space.

Prob: let $V = \mathbb{R}^2$ with addition defined as $(x,y)+(x',y') = (x+x'+1, y+y')$ is it vector space?

Sol: NO, V is not a vector space since additive identity does not exist.

i.e. Suppose $(x,y) \in \mathbb{R}^2$

Now $(x,y)+(0,0) = (x+1, y) \neq (x,y)$.

Prob: Is the set $V = \{ f : \mathbb{R} \to \mathbb{R} \mid f(0) = 1 \}$ a vector space

Solution. Not a vector space, because it is not closed under addition of two functions f & g.

i.e., $f, g \in V$.

then $(f+g)(0) = f(0) + g(0) = 1+1 = 2 \neq 1$.

Prob: let V be the set of all 2×2 real matrix show V is a vector space.

Prob: Is set of real numbers $\mathbb{R}$ is a vector space over $\mathbb{C}$

NO, $\mathbb{R}$ over $\mathbb{C}$ is not a vector space because Scalar Multiplication does not exist

i.e., $1 \in \mathbb{R}$ and $i \in \mathbb{C}$

the mapping $f \times V \to V$ field × Vector → Vector

$1 \cdot i = i \notin \mathbb{R}$.

here field is $\mathbb{C}$
Vector is $\mathbb{R}$.

Prob: $V = \{f : \mathbb{R} \to \mathbb{R} \mid f(x) = -f(x)\}$ Set of all functions is a vector space over $\mathbb{R}$.

Yes, V is a vector space because V is closed under addition and Scalar Multiplication.

i.e., let $f, g \in V$ and $a \in \mathbb{R}$.

$$(f+g)(-x) = f(-x) + g(-x) = -f(x) - g(x)$$
$$= -[f(x) + g(x)].$$
$$= -(f+g)(x).$$

Scalar Multiplication

$$(af)(-x) = af(-x) = a \times -f(x) = -af(x)$$

$\Rightarrow$ V is closed under addition and Scalar multiplication and also V satisfies all axioms of Vector Space [Exercise for students].

Hence, $V = \{f : \mathbb{R} \to \mathbb{R} \mid f(x) = -f(x)\}$ is a vector Space over $\mathbb{R}$.

Prob: Let $V = \mathbb{R}^2$ with a scalar multiplication defined as $a(x,y) = (ax, 0)$ is it a vector space?

Let $(x,y)\ (x'y') \in \mathbb{R}^2$

now $a[(x,y) + (x'y')] = a(x+x', y+y')$.

$= a[(x+y), + (x'y')]$

$= [a(x+x'), 0] \neq a(x;y) + a(x'y')$.

distributive Law over addition is not satisfied.

Hence $V = \mathbb{R}^2$ with scalar multiplication defined as $a(x,y) = (ax, 0)$.

Prob: Let $V = \{x \in \mathbb{R}^3 / Ax = 0\}$, where A is mxn matrix is a vector space?

In Exercises 1–16 a set of objects is given, together with operations of addition and scalar multiplication. Determine which sets are vector spaces under the given operations. For those that are not vector spaces, list all axioms that fail to hold.

1. The set of all triples of real numbers (x, y, z) with the operations

$(x, y, z) + (x', y', z') = (x + x', y + y', z + z')$ and $k(x, y, z) = (kx, y, z)$

2. The set of all triples of real numbers (x, y, z) with the operations

$(x, y, z) + (x', y', z') = (x + x', y + y', z + z')$ and $k(x, y, z) = (0, 0, 0)$

3. The set of all pairs of real numbers (x, y) with the operations

$(x, y) + (x', y') = (x + x', y + y')$ and $k(x, y) = (2kx, 2ky)$

4. The set of all real numbers x with the standard operations of addition and multiplication.

5. The set of all pairs of real numbers of the form $(x, 0)$ with the standard operations on R^2.

6. The set of all pairs of real numbers of the form (x, y), where $x \geq 0$, with the standard operations on R^2.

7. The set of all n-tuples of real numbers of the form $(x, x, \ldots, x)$ with the standard operations on R^n.

8. The set of all pairs of real numbers (x, y) with the operations

$(x, y) + (x', y') = (x + x' + 1, y + y' + 1)$ and $k(x, y) = (kx, ky)$

9. The set of all 2×2 matrices of the form

$$\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$$

with the standard matrix addition and scalar multiplication.

10. The set of all 2×2 matrices of the form

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

with the standard matrix addition and scalar multiplication.

11. The set of all real-valued functions f defined everywhere on the real line and such that $f(1) = 0$, with the operations defined in Example 4.

12. The set of all 2×2 matrices of the form

$$\begin{bmatrix} a & a+b \\ a+b & b \end{bmatrix}$$

with matrix addition and scalar multiplication.

13. The set of all pairs of real numbers of the form $(1, x)$ with the operations

$(1, y) + (1, y') = (1, y + y')$ and $k(1, y) = (1, ky)$

14. The set of polynomials of the form $a + bx$ with the operations
$(a_0 + a_1x) + (b_0 + b_1x) = (a_0 + b_0) + (a_1 + b_1)x$ and $k(a_0 + a_1x) = (ka_0) + (ka_1)x$

15. The set of all positive real numbers with the operations
$x + y = xy$ and $kx = x^k$

16. The set of all pairs of real numbers (x, y) with the operations
$(x, y) + (x', y') = (xx', yy')$ and $k(x, y) = (kx, ky)$

17. Show that the following sets with the given operations fail to be vector spaces by identifying all axioms that fail to hold.

(a) The set of all triples of real numbers with the standard vector addition but with scalar multiplication defined by $k(x, y, z) = (k^2x, k^2y, k^2z)$.

(b) The set of all triples of real numbers with addition defined by $(x, y, z) + (u, v, w) = (z + w, y + v, x + u)$ and standard scalar multiplication.

(c) The set of all 2×2 invertible matrices with the standard matrix addition and scalar multiplication.

18. Show that the set of all 2×2 matrices of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with addition defined by $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} + \begin{bmatrix} c & 1 \\ 1 & d \end{bmatrix} = \begin{bmatrix} a + c & 1 \\ 1 & b + d \end{bmatrix}$ and scalar multiplication defined by $k\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} = \begin{bmatrix} ka & 1 \\ 1 & kb \end{bmatrix}$ is a vector space. What is the zero vector in this space?

19.

(a) Show that the set of all points in R^2 lying on a line is a vector space, with respect to the standard operations of vector addition and scalar multiplication, exactly when the line passes through the origin.

(b) Show that the set of all points in R^3 lying on a plane is a vector space, with respect to the standard operations of vector addition and scalar multiplication, exactly when the plane passes through the origin.

20. Consider the set of all 2×2 invertible matrices with vector addition defined to be matrix *multiplication* and the standard scalar multiplication. Is this a vector space?

21. Show that the first nine vector space axioms are satisfied if $V = R^2$ has the addition and scalar multiplication operations defined in Example 5.

22. Prove that a line passing through the origin in R^3 is a vector space under the standard operations on R^3.